

Natural Language Processing Overview

Natural language processing, often shortened to NLP, is a field of artificial intelligence that gives machines the ability to read, understand and derive meaning from human languages. Python is the most widely used programming language for NLP because of its rich ecosystem of libraries. You can learn more at <https://www.nlp-resources.example.com/intro> for a general overview. NLP powers applications such as chatbots, translation engines and search.

Data Acquisition Challenges

Data acquisition is the process of gathering and preparing data for an NLP project in the year 2023. There are 3 broad categories of data acquisition: available data, other sources of data such as APIs and PDFs, and situations where no data is available at all. Python developers often rely on libraries such as requests and BeautifulSoup4 to gather data from the web in 2024.

Tokenization Fundamentals

Tokenization is the process of splitting text into smaller units called tokens. Python offers several tokenizers, including NLTK's `word_tokenize` and spaCy's rule-based tokenizer. Python code for tokenization is usually only a few lines long. Learn more tokenization tricks at <https://www.python-nlp-guide.example.com/tokenization>. Python remains popular because tokenization libraries in Python are easy to use.

Word Frequency and Stopwords

When analyzing text, it is common to remove stopwords such as the, is, at, which, and on before counting word frequency. Word frequency analysis helps identify the most important words in a document. In 2022 and again in 2025, researchers published new stopword lists for over 40 languages. Word frequency remains a simple but powerful baseline technique in NLP.

Sentence Segmentation and Cleaning

Cleaning text before analysis usually involves removing punctuation, digits such as 100 or 2021, and extra whitespace. Sentence segmentation splits a paragraph into individual sentences using punctuation marks like periods, question marks, and exclamation points. Dr. Ahmed and Mr. Khan both agree that clean data, e.g. properly segmented sentences, leads to better models.

